

Two Electrons on Helium as a Differential Sensor for Majorana Zero Modes

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We propose using a pair of Coulomb-entangled electrons on superfluid helium, prepared in the motional/spin singlet configuration used as a differential field sensor in Ref. [?], to probe local signatures associated with Majorana zero modes (MZMs) in a proximal topological superconductor. Because the singlet is insensitive to common-mode perturbations and maximally sensitive to a *difference* between the two trap sites, positioning the two electrons asymmetrically with respect to a topological wire converts a local, sub-shot-noise MZM signature into a differential signal read out through the same quantum-Fisher-information framework already developed for particle-physics sensing on this platform. This is a proposal: we outline the signal channels under consideration, the required theoretical extension of the existing singlet–triplet sensor model, and the open questions that must be resolved before a concrete protocol can be simulated.

I. INTRODUCTION

Two electrons on superfluid helium, entangled through their unscreened Coulomb interaction, have been shown theoretically to support a differential-sensing scheme: prepared in the motional/spin singlet, the pair is blind to common-mode fields but maximally responsive to a *gradient* between the two trap sites, quantified through the quantum and classical Fisher information and benchmarked against an independent-spin Ramsey scheme [?]. That proposal targeted generic magnetic-field gradients (e.g. from light dark matter or axion-like fields). The underlying entangled states, and the machinery for computing them (Hartree/CI on a sinc-DVR basis, configurations I/II/III), were established for the two-electron helium platform in Ref. [?], with time-dependent control of the same states realized as high-fidelity two-qubit gates in Ref. [?]. Reference [?] also grounds the sensing readout of a real device in Refs. [? ?], which demonstrate the resonator-based charge/orbital readout hardware this platform actually has today.

Majorana zero modes (MZMs), predicted to occur at the ends of one-dimensional topological superconductors [?] and in proximitized semiconductor nanowires [? ?], have been pursued experimentally for over a decade, with zero-bias conductance signatures reported in hybrid superconductor–semiconductor nanowires [?] and reviewed extensively since [? ?]. A central experimental difficulty is that MZM signatures are subtle, easily mimicked by trivial (non-topological) Andreev bound states, and sensitive to quasiparticle poisoning — a stray quasiparticle tunneling onto the topological segment flips its fermion parity and can toggle any locally coupled signal. This is precisely the kind of weak, potentially transient, spatially localized perturbation that a differential entangled-pair sensor is suited to pick up while rejecting the (much larger) common-mode electromagnetic environment of a cryostat.

This project asks whether the existing eHe differential-sensor machinery can be repurposed, with the “target signal” reinterpreted from a magnetic-field gradient to

whatever local field or charge signature a nearby MZM produces. We do not yet have a settled answer to *which* physical channel (Sec. II) is both (i) strong enough to be detectable at achievable electron–source distances and (ii) compatible with the existing singlet–triplet coupling operator. The goal of this manuscript, at this stage, is to lay out the extension of the model and the open questions rather than to report a result.

II. MODEL

A. Reusing the existing sensor Hamiltonian

The starting point is the effective two-level (ground/excited, singlet-like/triplet-like) model of Ref. [?]:

$$H_{\text{eff}} = E_g |g\rangle\langle g| + E_e |e\rangle\langle e| + \lambda(t) (|g\rangle\langle e| + |e\rangle\langle g|), \quad (1)$$

where in the original proposal $\lambda = (g\mu_B\delta B/2)M_{ge}$ couples to a magnetic-field gradient $\delta B = B_L - B_R$ via the matrix element $M_{ge} = \langle e|S_L^z - S_R^z|g\rangle$. For a Majorana-mode signal, $\lambda(t)$ should instead encode whatever locally asymmetric quantity the MZM produces at the two trap sites — to be determined (see below). The quantum/classical Fisher information and Cramér–Rao bound machinery of Ref. [?] carries over unchanged once $\lambda(t)$ is specified.

B. Candidate signal channels (to be evaluated)

- **Parity-switching noise.** A quasiparticle-poisoning event on the topological segment flips the ground-state fermion parity, which (depending on the specific hybrid architecture) can modulate a local charge or field asymmetric between the two trap sites. This would appear as a telegraph-noise-like signal rather than a coherent Rabi oscillation, requiring a different Fisher information treatment

(parameter = switching rate, not a static/slowly-varying field). This picture is no longer purely hypothetical: a minimal (two-site) Kitaev-chain realization of an MZM pair has now had its joint parity read out in real time via quantum capacitance, directly observing random-telegraph parity switching with lifetimes exceeding 1 ms [?]; that measurement is an intrinsic, local (dispersive) probe of the topological device itself, not an external differential sensor, so it grounds the *statistics* of the target signal (telegraph process, not a static field) without yet demonstrating anything like the remote, common-mode-rejecting readout proposed here. Independently, measured quasiparticle-poisoning statistics in conventional superconducting qubits (correlated multi-qubit bursts, a power-law-decaying event rate from phonon-only sources) offer a separate, concrete noise-process model to benchmark a telegraph-process Fisher-information treatment against [?]. The closest existing precedent to this project’s overall detection philosophy — using a qubit’s parity flip as a rare-event particle detector — is the quantum-parity-detector (QPD) concept, which uses a single superconducting qubit’s own quasiparticle parity (not an entangled pair, and not an MZM) for meV-threshold dark-matter/rare-event searches [?]; the entangled, common-mode-rejecting, MZM-specific angle pursued here appears not yet covered by that or any other proposal found in this project’s literature search.

- **Stray field from Majorana-induced edge/Andreev currents.** If the topological wire supports a weak stray magnetic field at its end (from the associated Andreev bound state charge distribution or edge current), this maps most directly onto the existing δB channel and requires the smallest change to the Ref. [?] formalism.
- **Direct charge-coupling channel.** Reusing the electric susceptibility / dispersive framework of Ref. [?] (an electron’s motional state already couples to local charge/potential fluctuations through the resonator), a nearby MZM’s contribution to the local electrostatic potential could couple directly to $(n_L - n_R)$ rather than to spin — a fundamentally different (charge, not spin-gradient) sensing channel that would need its own generator and its own QFI expression.

Each channel implies a different coupling operator, a different noise model, and a different distance/coupling-strength budget; identifying the most promising one (or ruling all three out at currently achievable electron–nanowire separations) is the first concrete task for this project.

C. Open theoretical questions

- What is the realistic magnitude of the candidate signal at achievable electron–MZM separations, and how does it compare to the δB magnitudes assumed in Ref. [?]?
- Is the relevant noise process (e.g. quasiparticle poisoning) better modeled as a static parameter to estimate (Fisher information / Cramér–Rao, as in the existing framework) or as a stochastic process requiring a different estimation-theoretic treatment (e.g. change-point detection, quantum non-demolition parity readout)? The direct, real-time observation of random-telegraph parity switching in a minimal Kitaev chain [?] answers this for the *intrinsic* parity-readout case (yes, a stochastic telegraph process, not a static field) — but this project’s proposed external, differential readout of a signal *sourced by* an MZM’s parity flip, rather than a direct probe of the MZM itself, may still see an effectively different (e.g. filtered, distance-attenuated) process, so the estimation-theoretic treatment cannot simply be copied over without re-deriving it for the external-sensor geometry.
- Does the factor-of-2 QFI advantage carry over quantitatively, or does the new coupling operator’s matrix element M_{ge} (or its charge-channel analogue) reduce the achievable gain?
- What does the required electron–nanowire integration look like physically (shared substrate? proximity-coupled chip?), and is it compatible with the trap/reservoir geometry of Ref. [?]?

III. RESULTS

Not yet available — this section will report the candidate-channel comparison and, for the strongest candidate, a QFI/CRB estimate analogous to Sec. IV of Ref. [?].

IV. DISCUSSION

If a viable signal channel is identified, next steps are: (i) extend the CI/Hartree machinery of Ref. [?] to include the new coupling operator, reusing the numerical toolkit described in the project’s `ci-dvr-hartree-numerics` skill; (ii) simulate the achievable QFI as a function of electron–MZM distance and compare against the independent-spin (Ramsey) benchmark; (iii) assess whether RL-based trap optimization (see the companion `rl-agentic-quantum-control` project) can improve the achievable coupling by reshaping the trap potential, as it already does for the field-gradient sensor.

V. CONCLUSIONS AND PERSPECTIVES

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ACKNOWLEDGMENTS

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